# Off-street parking on rue de Terrebonne as built: a reading of the 1954–55 fire-insurance plans

Terrebonne-open-sources (a resident of Côte-des-Neiges–Notre-Dame-de-Grâce)

October 25, 2026 v2026-10-25-14-29-58

## Summary

Of the **126** residential buildings with a rue de Terrebonne address between the west end of the protected lane (Belmore) and avenue Girouard on the sixteen sheets available, **75** were built with a garage, that is **60 %** (north side 57 of 90, 63 %; south side 18 of 36, 50 %). Counting only the readings made with high confidence gives 62 %. Weighted by the dwellings the City's assessment roll lists in the same buildings, 59 % of the 324 dwellings are in a building that was built with a garage.

The plans show garages as built, not driveways: a driveway without a garage leaves no mark, so the share of properties with a driveway or a garage as built is at least this figure. They show the stock of 1954–55, not of today; the City's roll dates 118 of the 121 matched buildings to 1955 or earlier (median 1932), so the frontage drawn is essentially the one standing today, but the use of each garage and driveway since then is not recorded here.

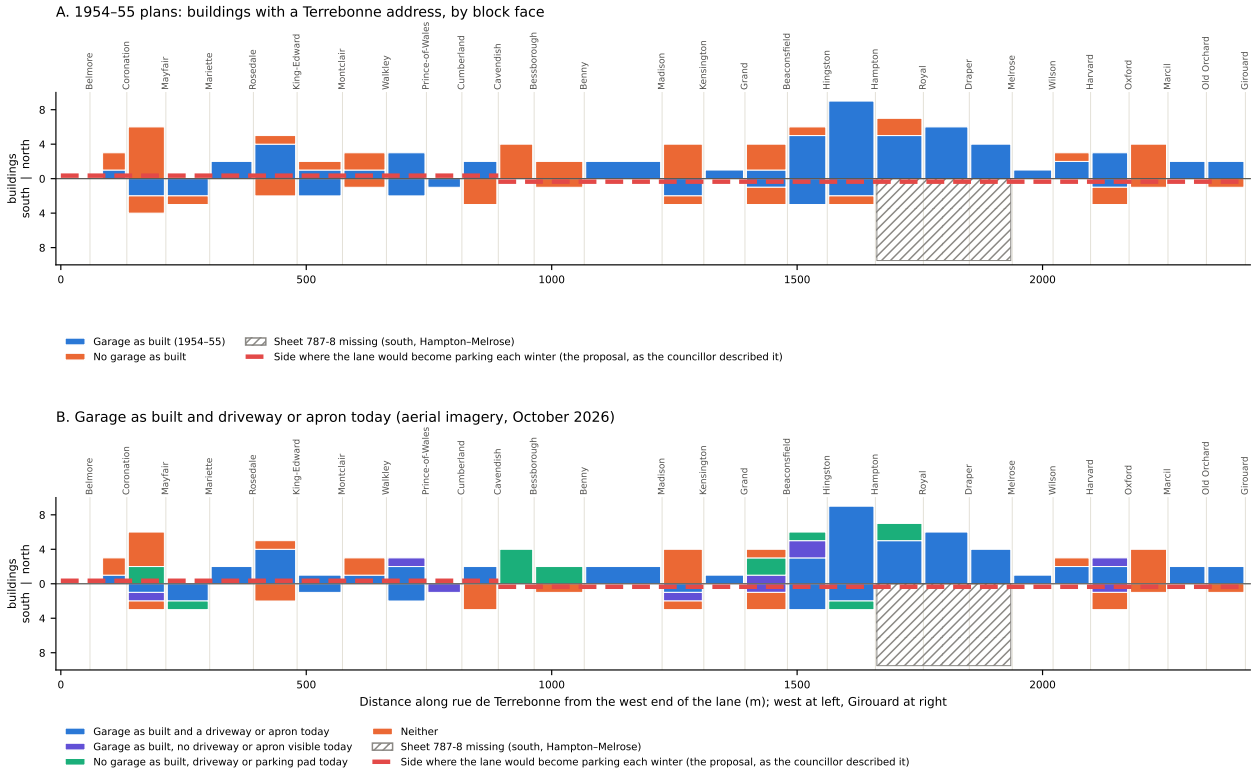



Figure 1: Off-street parking along the whole street, block face by block face (north side above the axis, south side below; west at left). A: the 1954–55 reading. B: the same buildings with the present-day reading. The dashed red line marks the side where the lane is to be converted to parking each winter under the proposal, as the councillor described it (the lane running with traffic: north side west of Cavendish, south side east of it).




Figure 2: Overview map, west half: garage as built (1954–55) and driveway or apron today, one marker per building, on the City's parcels (continued on the next page; legend at the end).

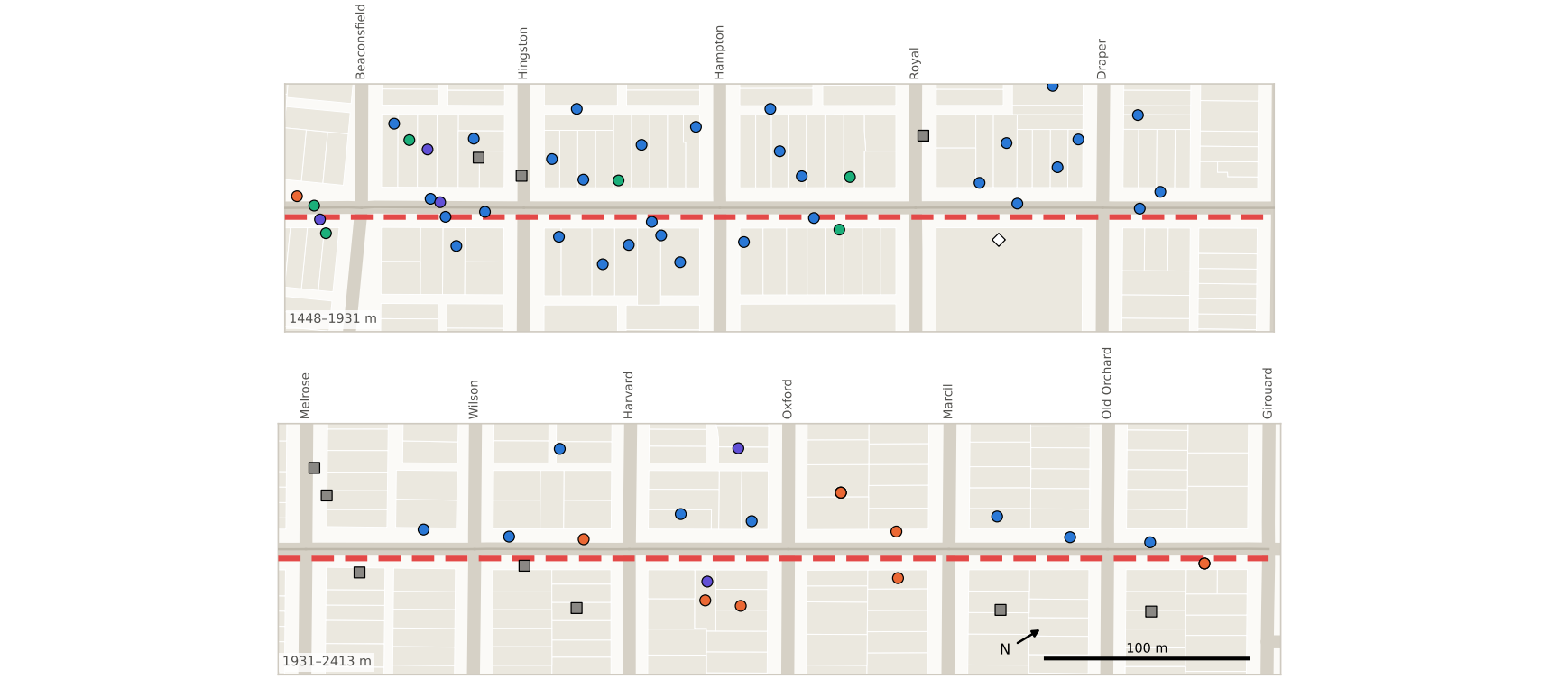



Figure 3: Overview map, east half, and the legend.

# 1 Source

The Underwriters' Survey Bureau, Limited, *Insurance plan of the city of Montreal*, volume 7, sheets 739-2 to 739-5, 775, 784-1, 784-2, 785-1, 787-5 to 787-7, 789-12 to 789-16 and 798-3, dated November 1954 to December 1955 on the sheets ("Montreal, vol. 7, November 1954"; sheet 775 "April 1955"; sheet 739-4 "December 1955"). Each sheet carries the notice "This plan is the property of the Underwriters' Survey Bureau, Limited, (unauthorized reproduction prohibited) . . . " and "Copyright registered at Ottawa by the Underwriters' Survey Bureau, Limited, Toronto and Montreal". The plans draw every building with its construction, storeys and civic number, and write *Auto* or *Autos* on every garage: in the basement (*Auto Bt*, *Autos Bast*), attached, or detached in the yard. The scans come from Bibliothèque et Archives nationales du Québec (BAnQ numérique, collection *Plans de villes et villages du Québec*, notice 0000174390, 273 files, `https://collections.banq.qc.ca/ark:/52327/2244199`; rights statement *Copyright Not Evaluated*); the full sheets are consulted there and not copied here. Sheet 787-8 (south side, Hampton to Melrose: the Royal and Draper blocks) is absent from BAnQ's file list (787-1 to 787-7 only) and from every other copy found; those two block faces are not counted (the north side opposite them has 15 garages on 17 buildings).

# 2 Method

1. **Reading.** Every building on the Terrebonne block faces, both sides, was read from the scans at two to four times magnification and recorded in `buildings_1954.csv` (one row per building: sheet, side, block, cross streets, civic numbers, building type, the street it is addressed on, corner lot, garage yes/no, kind of garage, the presumed access, a confidence grade and notes). The unit is the building with a Terrebonne civic number: a semi-detached pair is two buildings, a duplex pair is one building of two to four flats, an apartment building counts once. Corner buildings addressed on the cross street (28, of which 13 with a garage) are listed with the cross street as their frontage and left out of the share; so are the two churches, the presbytery and the two frontages vacant in 1954–55.

2. **Georeferencing.** Each sheet was fitted to the City's géobase (CC-BY 4.0) by a similarity transform on the pixel positions of its Terrebonne intersections (`controls.csv`, `georef.py`); residuals are 0–4 m (one of 7 m). The fit was checked by drawing the City's assessment-roll parcels over each sheet.

3. **Positions and dwellings.** Each row was joined to the City's *unités d'évaluation foncière* (CC-BY 4.0) by civic number and street, or, for corner buildings renumbered on the cross street since 1954, by position (`buildings.py`; 150 of 161 rows matched, the rest placed schematically along their block face). The roll gives the dwelling count and the year of construction of each matched building.

4. **Maps.** Both maps use a frame straightened along the street (distance along Terrebonne, offset across; north side above). The first draws the markers on the georeferenced sheets, the second on a base made only of the City's open data (`maps.py`).

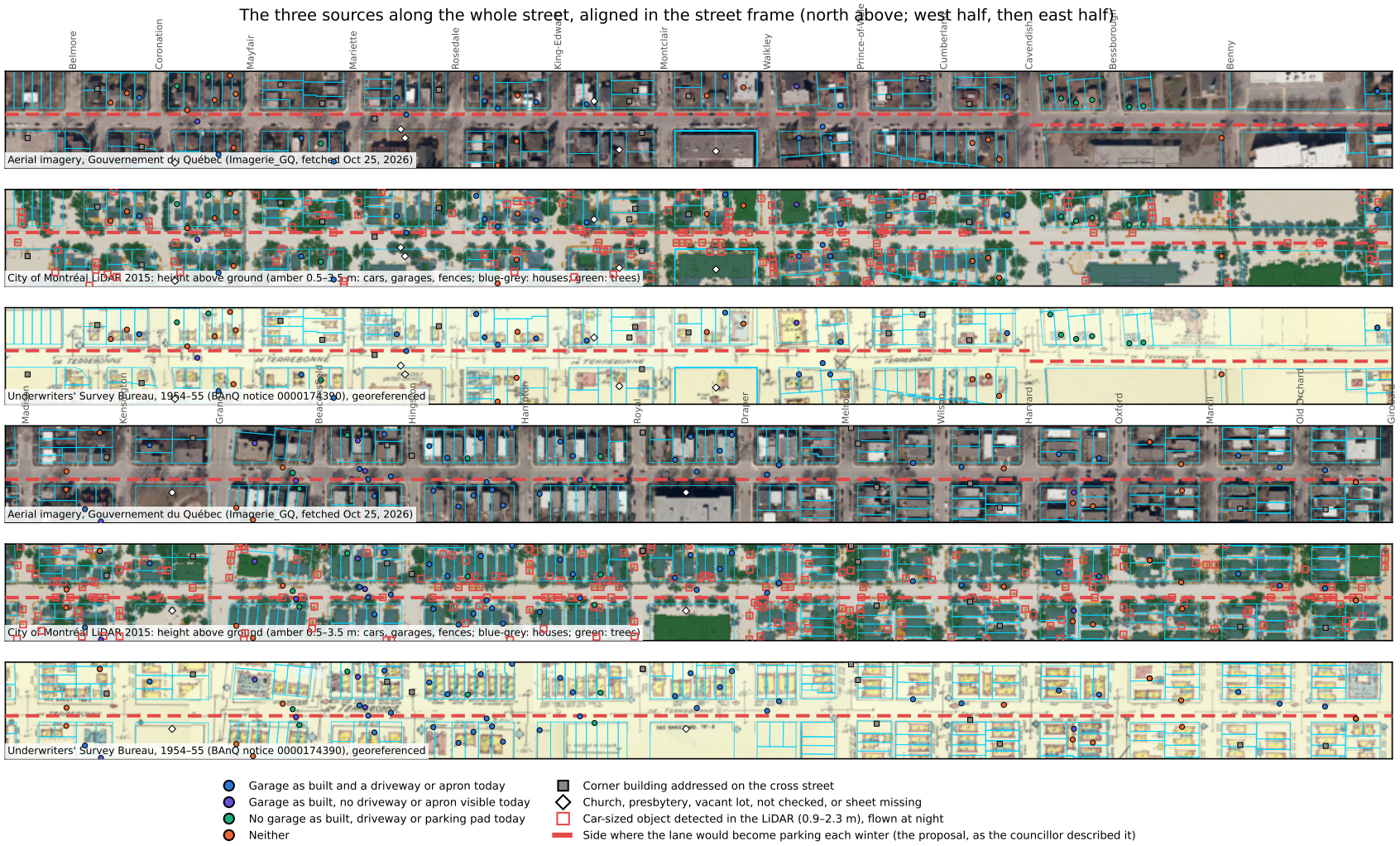
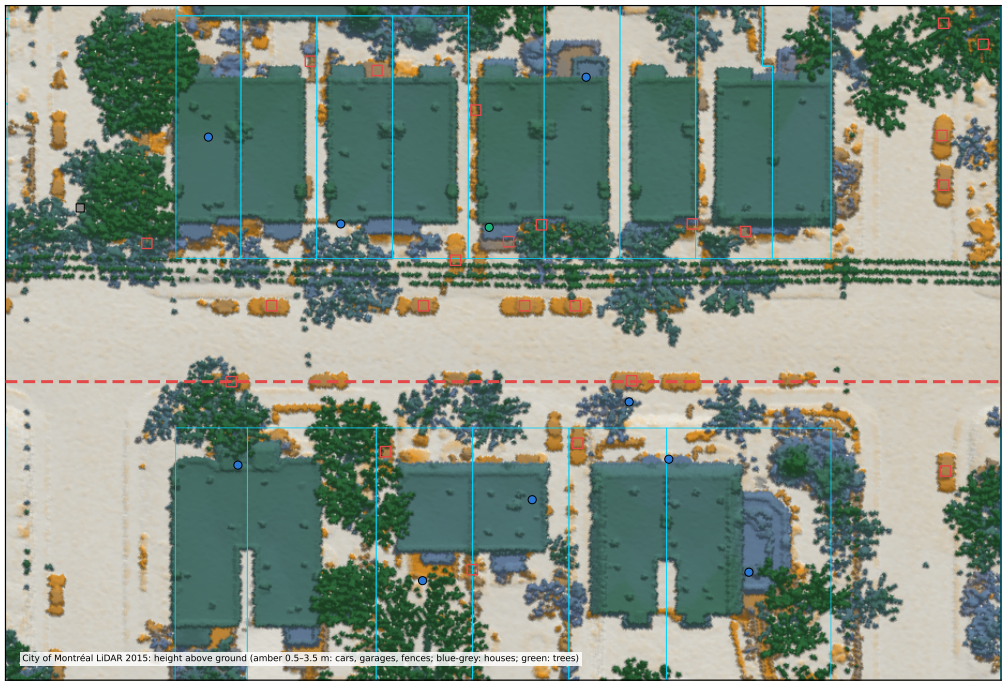
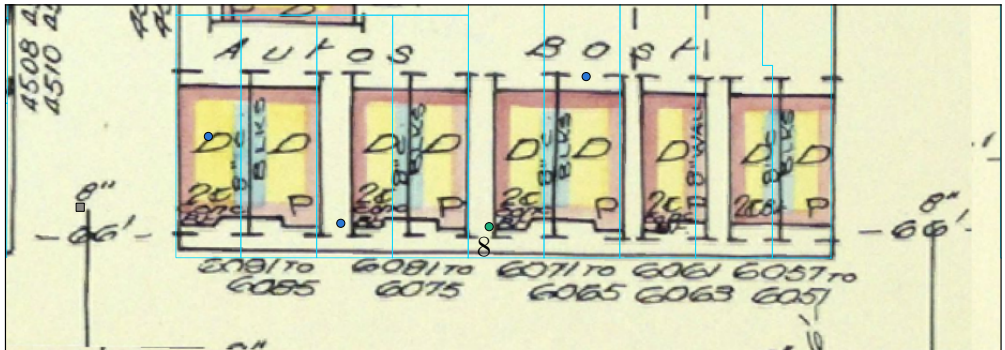



Figure 4: How the three sources were used, whole street: for each half of the street, the Gouvernement du Québec aerial imagery, the City's 2015 LiDAR as height above ground, and the 1954–55 sheets, all georeferenced into the same street frame, with the building markers.

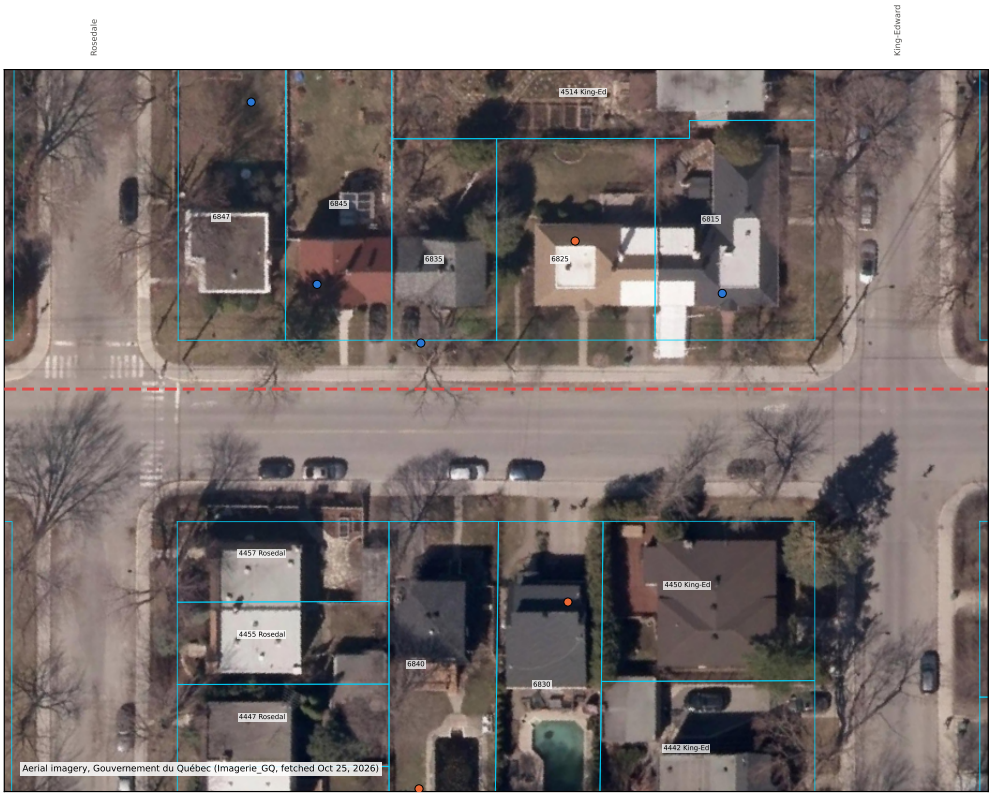
One block, Hingston to Hampton: aprons and parked cars (optical), the same cars and the garages as 0.5–3.5 m objects (LiDAR), the « Autos Bast » marks of 1954 (sheet 787-5 north, 787-7 south)

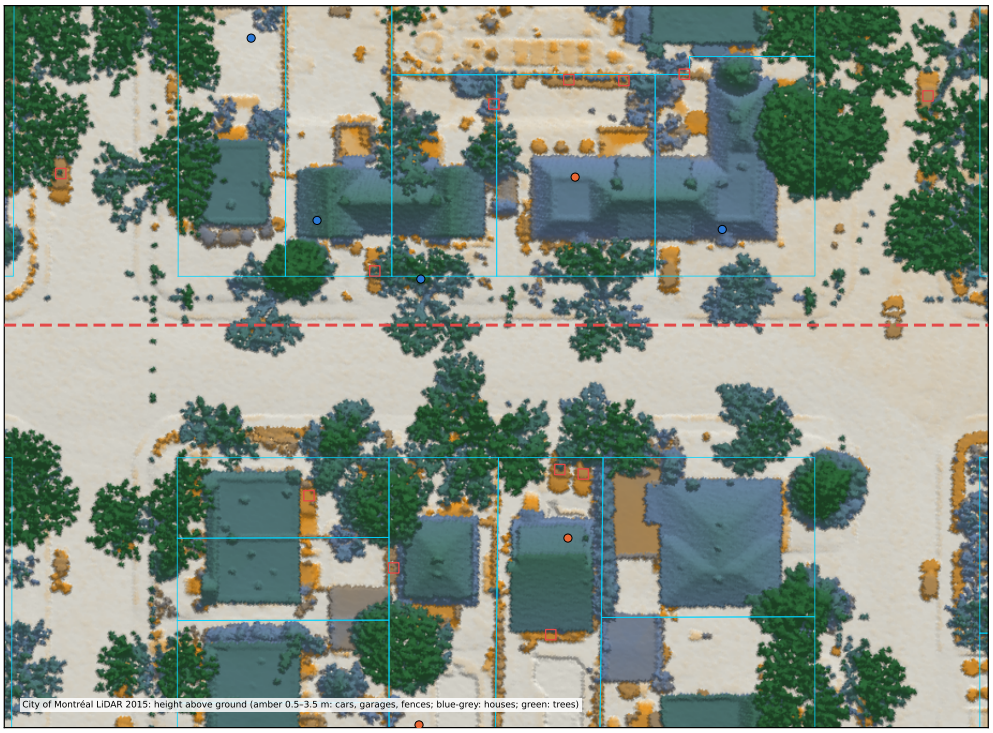


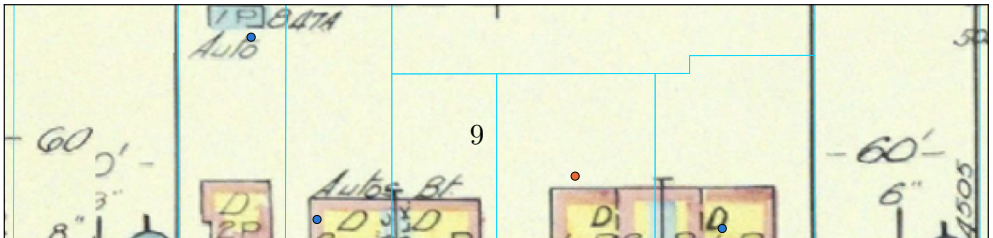



One block, Rosedale to King-Edward: detached rear garages of 1954 (sheet 789-14) seen in the LiDAR as 2.5–3.5 m boxes; aprons and a parked car on the optical image



# 3 Results

| | Buildings addressed on Terrebonne | With a garage as built | Share |
|---|---|---|---|
| North side | 90 | 57 | 63 % |
| South side (sheet 787-8 missing) | 36 | 18 | 50 % |
| **Both sides** | **126** | **75** | **60 %** |
| High-confidence readings only | 97 | 60 | 62 % |
| Dwellings (City roll, 121 buildings) | 324 | 192 | 59 % |

Table 1: Garages as built on the Terrebonne frontage, 1954–55 plans.

Of the 75 garages, 46 are basement garages, 14 are detached in the yard and 15 attached or in an annex. The door opens on Terrebonne for 64, on either street for 10 corner lots, and on a rear lane for 1. Building types on the frontage (from the plans): semi 38, single 15, duplex 67, row 2, apartment 4. Years of construction from the roll, by decade: 1910s 1, 1920s 29, 1930s 44, 1940s 15, 1950s 29, 1960s 1, 2000s 2.

Table 2: By block face (residential buildings addressed on Terrebonne).

| Side | Sheet | Block face | Buildings | Garage | Share |
|---|---|---|---|---|---|
| N | 789-12 | Belmore-Coronation | 3 | 1 | 33 % |
| N | 789-13 | Coronation-Mayfair | 6 | 0 | 0 % |
| N | 789-13 | Mariette-Rosedale | 2 | 2 | 100 % |
| N | 789-14 | Rosedale-King Edward | 5 | 4 | 80 % |
| N | 789-14 | King Edward-Montclair | 2 | 1 | 50 % |
| N | 789-14 | Montclair-Walkley | 3 | 1 | 33 % |
| N | 789-15 | Walkley-Prince of Wales | 3 | 3 | 100 % |
| N | 789-15 | Cumberland-Cavendish | 2 | 2 | 100 % |
| N | 789-16 | Cavendish-Bessborough | 4 | 0 | 0 % |
| N | 789-16 | Bessborough-Benny | 2 | 0 | 0 % |
| N | 798-3 | Benny-Madison | 2 | 2 | 100 % |
| N | 798-3 | Madison-Kensington | 4 | 0 | 0 % |
| N | 798-3 | Kensington-Grand | 1 | 1 | 100 % |
| N | 787-5 | Grand-Beaconsfield | 4 | 1 | 25 % |
| N | 787-5 | Beaconsfield-Hingston | 6 | 5 | 83 % |
| N | 787-5 | Hingston-Hampton | 9 | 9 | 100 % |
| N | 787-6 | Hampton-Royal | 7 | 5 | 71 % |
| N | 787-6 | Royal-Draper | 6 | 6 | 100 % |
| N | 787-6 | Draper-Melrose | 4 | 4 | 100 % |
| N | 739-2 | Melrose-Wilson | 1 | 1 | 100 % |
| N | 739-2 | Wilson-Harvard | 3 | 2 | 67 % |
| N | 739-2 | Harvard-Oxford | 3 | 3 | 100 % |
| N | 739-5 | Oxford-Marcil | 4 | 0 | 0 % |
| N | 739-5 | Marcil-Old Orchard | 2 | 2 | 100 % |
| N | 739-5 | Old Orchard-Hampstead apts | 1 | 1 | 100 % |
| N | 739-5 | Old Orchard-Girouard | 1 | 1 | 100 % |
| S | 784-1 | Coronation-Mayfair | 4 | 2 | 50 % |
| S | 784-2 | Mayfair-Mariette | 3 | 2 | 67 % |

| Side | Sheet | Block face | Buildings | Garage | Share |
|---|---|---|---|---|---|
| S | 784-2 | Rosedale-King Edward | 2 | 0 | 0 % |
| S | 785-1 | King Edward-Montclair | 2 | 2 | 100 % |
| S | 785-1 | Montclair-Walkley | 1 | 0 | 0 % |
| S | 785-1 | Walkley-Prince of Wales | 2 | 2 | 100 % |
| S | 785-1 | Prince of Wales-Cumberland | 1 | 1 | 100 % |
| S | 785-1 | Cumberland-Cavendish | 3 | 0 | 0 % |
| S | 775 | Cavendish-Madison | 1 | 0 | 0 % |
| S | 775 | Madison-Kensington | 3 | 2 | 67 % |
| S | 787-7 | Grand-Beaconsfield | 3 | 1 | 33 % |
| S | 787-7 | Beaconsfield-Hingston | 3 | 3 | 100 % |
| S | 787-7 | Hingston-Hampton | 3 | 2 | 67 % |
| S | 739-4 | west cross street-Oxford | 3 | 1 | 33 % |
| S | 739-5 | Oxford-Marcil | 1 | 0 | 0 % |
| S | 739-5 | Old Orchard-Girouard | 1 | 0 | 0 % |

Belmore
Coronation
Mayfair
Mariette
Rosedale
King-Edward
0–483 m
King-Edward
Montclair
Walkley
Prince-of-Wales
Cumberland
Cavendish
Bessborough
483–965 m
Bessborough
Benny
Madison
Kensington
Grand




Figure 8: The 1954–55 reading on the City's parcels, east half, and the legend.

Belmore
Coronation
Mayfair
Mariette
Rosedale
King-Edward
DE TERREBONNE
DE TERREBONNE
0–483 m
King-Edward
Montclair
Walkley
Prince-of-Wales
Cumberland
Cavendish
Bessborough
DE TERREBONNE
DE TERREBONNE
8570
483–965 m
Bessborough
Benny
Madison
Kensington
Grand

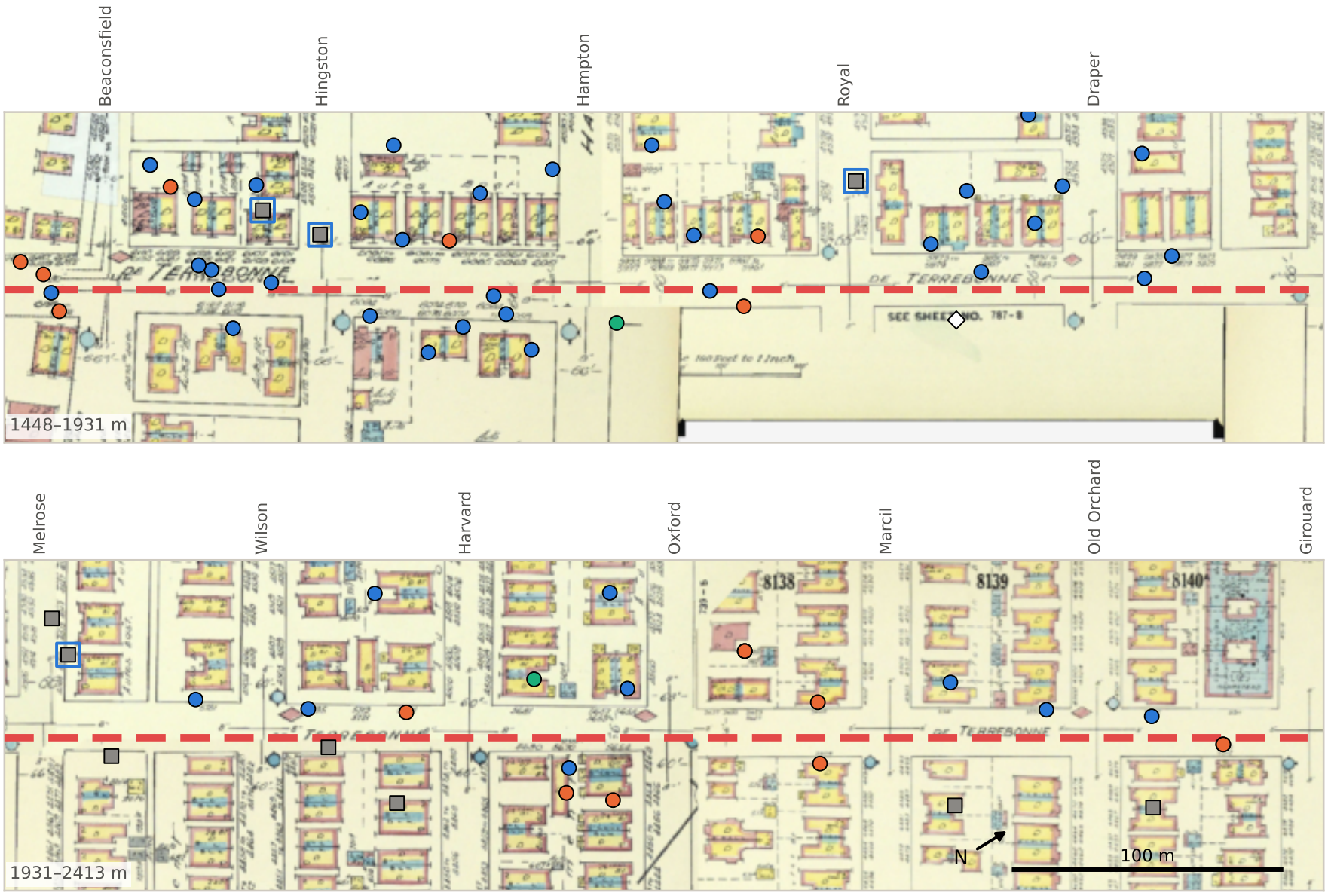



Figure 10: The 1954–55 sheets, east half, and the legend.

# 4 The frontages today

A check of each building's frontage on the Gouvernement du Québec's continuous aerial imagery (service *Imagerie_GQ*, fetched October 25, 2026; leaf-off spring imagery, about 8 cm per pixel; the City's parcels overlaid) recorded, for 123 of the 126 buildings, whether a driveway, paved apron or parking pad is visible on the lot today, whether a vehicle was parked on it at the moment of the photograph, and from which street it is reached (`buildings_geo.csv`, columns `driveway_today`, `vehicle_or_pad_on_lot`, `access_today`). What a garage holds inside cannot be seen from above, so the check measures the visible access, not the use.

| | Buildings | Share |
|---|---|---|
| Checked | 123 | |
| Driveway, apron or parking pad visible today | 79 | 64 % |
| Garage as built (1954–55) *or* driveway today | 89 | 72 % |
| north side | 70 of 89 | 79 % |
| south side | 19 of 34 | 56 % |
| by dwellings (City roll) | 221 of 303 | 73 % |
| Garage of 1954–55 with its apron still visible | 64 | |
| Garage of 1954–55 with no apron visible today | 10 | |
| No garage in 1954–55, a driveway or pad today | 15 | |
| A vehicle parked on the lot when photographed | 34 | |

Table 3: The frontages today (aerial imagery read October 25, 2026).

# 5 Cars parked on the street in the two acquisitions

Both remote-sensing sources also show the cars parked on rue de Terrebonne itself, at two very different moments. The aerial imagery was taken in daytime, in a leaf-off spring, after June 2024 (the lane markings are visible); counting by eye on the block strips gives 158 cars at the Terrebonne curbs (north side 30, south side 128), of which 90 on the side where the lane would be converted. The City's LiDAR was flown in 2015, before the lane, at night (the GPS times put the corridor passes between 00:00 and 04:00 local, on two nights; the flight dates are not in the files; the trees are in leaf, so late spring to early autumn). Car-sized objects, 0.9–2.3 m above ground and 1.4–3 m wide, detected automatically in the point cloud (`lidar_cars.py`; rows of touching cars split by length), number 85 at the Terrebonne curbs (north 38, south 47), 253 on private lots within 45 m of the street (both sides, including lots of cross-street houses) and 70 on the cross streets within the same band. The night count is the one that bears on resident parking; in 2015 the street had parking on both sides (EXP counted 475 spaces in 2022 before the redesign, 171 after). The automatic count misses vehicles under tree crowns and tall vans, and may take a hedge or a low shed on a lot for a car, so the lot figure is an upper bound and the curb figures are the firmer ones. Table **??** and figure **??** give the block-by-block counts (`on_street_cars_by_block.csv`).

| Cars on rue de Terrebonne | North side | South side | Total |
|---|---|---|---|
| Aerial imagery, daytime, spring, after June 2024 | 30 | 128 | 158 |
| City LiDAR 2015, night (00:00–04:00), before the lane | 38 | 47 | 85 |
| LiDAR 2015, night: car-sized objects on private lots within 45 m | 144 | 109 | 253 |
| LiDAR 2015, night: on the cross streets within the same band | | | 70 |

Table 4: Cars parked at the moment of each acquisition.

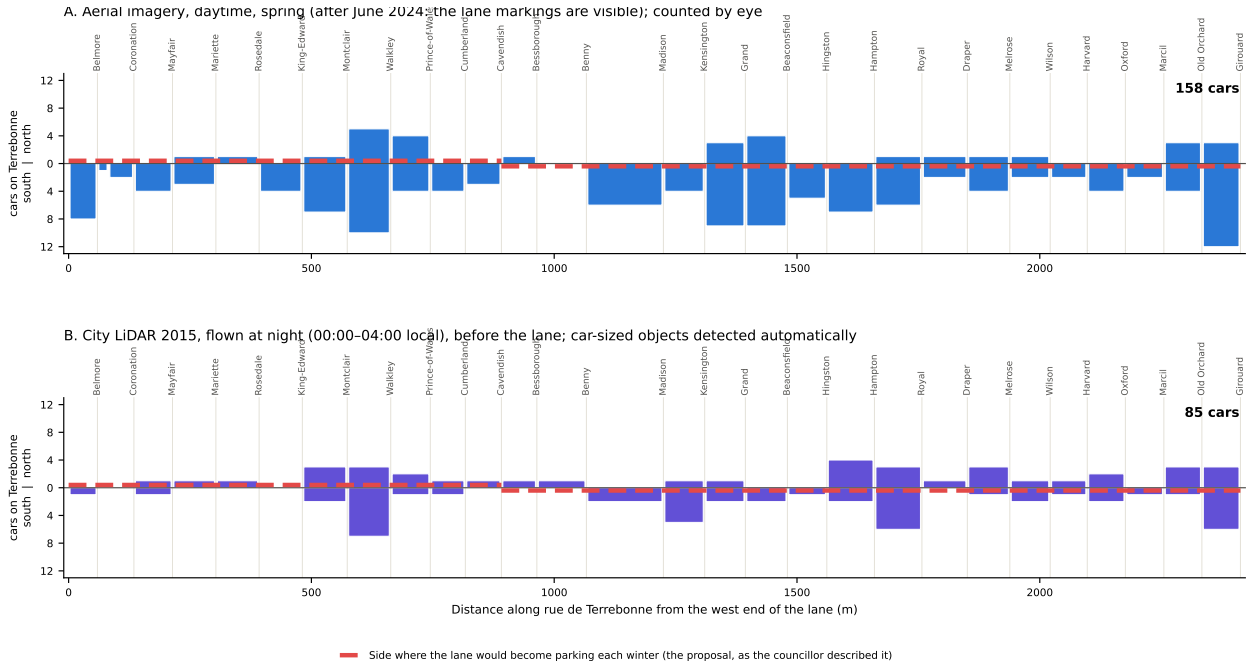



Figure 11: Cars parked on rue de Terrebonne, block by block and by side, in the daytime imagery (A) and in the 2015 night LiDAR (B).

## 6 Checks on the method

A reader who doubts a figure should be able to see what would have to be wrong for it to be wrong. The checks made, and what they found:

1. **Georeferencing.** Each sheet is fitted by a similarity transform (scale, rotation, translation) on its Terrebonne intersections; the fit is over-determined on fourteen of sixteen sheets. Residuals at the control points: mean 0.3–3.5 m per sheet, largest 7.0 m on sheet 739-4 (one intersection), 4.3 m on 787-7, under 3.3 m on every other sheet (`sheet_transforms.json`). The fit was then checked by drawing the City's parcel polygons over every sheet (`georef.py -check`): the 1954 lot lines and today's parcels coincide to within a lot width along the whole street. A residual of a few metres cannot move a building from one lot to the next, let alone from one side of the street to the other.

2. **The 1954–55 reading against an earlier, independent one.** In June 2024, before this question was put, the repository's author marked every *Auto* on the same sheets for another purpose (a band count, Somerled to Monkland). Those marks, carried in the table as `owner_star`, agree with the present building-by-building reading on all 126 buildings addressed on Terrebonne

(75 with, 51 without). The two readings were made two years apart, by different readers, the first without any per-building or per-side purpose.

3. **The present-day reading.** It is a single reader's reading of public imagery, graded row by row (high, medium, unclear) with a note on what was seen; the strips with the parcels overlaid are regenerated by `ortho_strips_qc.py` from the public service, so any row can be re-read. It was made, and the per-side tallies computed, before the conversion side was drawn on any figure.

4. **The LiDAR car detector.** On four blocks (45 car-shaped objects at the Terrebonne curbs counted by eye on the height rasters), the detector found 34 (`lidar_detector_check.csv`): about three quarters, the misses being cars under tree crowns and rows of touching cars split imperfectly. The LiDAR curb count of 85 is therefore a lower bound, roughly 110 after correcting for recall; it is reported as detected, with the correction stated here and not applied in the tables.

5. **Join to the roll.** 150 of 161 rows match a City assessment unit by civic number and street, or by position for corners renumbered since 1954; the 11 unmatched (cross-street corner groups, the missing sheet, two vacant frontages, two buildings since replaced) are placed schematically and carry no dwelling count. The dwelling-weighted shares use the matched rows only.

6. **A census, not a sample.** Every building with a Terrebonne address between the west end of the lane and Girouard on the sheets available is in the table; nothing was sampled or extrapolated except the recall correction above, which is not applied.

7. **Reproducibility.** Every number is produced by a script in `derived/` from inputs that are public: the City's open data (CC-BY 4.0), the Gouvernement du Québec's imagery service, and BAnQ's copy of the plans. The only hand-made inputs are the two readings, each a CSV with one row per building.

# 7 Caveats

- **Garages, not driveways.** The plans mark structures. A driveway or parking pad without a garage leaves no mark, so the as-built share with a driveway or garage is a lower bound.

- **As built, not as used.** The 1954–55 share counts structures; the present-day check sees aprons, driveways and parked vehicles from above but not what a garage holds inside (a workshop, storage, living space), so neither figure is the share of households that park off-street today. A garage built in 1930 or 1950 may also be too small for a large vehicle of today. Ten garages of 1954–55 show no apron today and one reading is unclear.

- **Buildings, not households.** The share by building and the share by dwelling happen to agree (60 % and 59 %), but a duplex pair has two to four flats and one or two garages, and three of the four apartment buildings have a basement garage of unrecorded capacity.

- **One sheet missing** (787-8) and 29 readings graded medium (noted row by row).

- **Two moments, not a time series.** The daytime imagery and the night LiDAR are ten years and a redesign apart; their car counts describe each moment and are not before-and-after figures for the lane.

- **Corner lots.** Which street a corner building is addressed on changed in places between 1954 and the present roll; the table keeps the 1954 numbering and the join records the present unit.